

MATH 1342 – Elementary Statistics – Week 9: P-values and Statistical Significance

Oct 20 – 22 | 4.2 Measuring Evidence with p-values • 4.3 Determining Statistical Significance | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
3. Find and use empirical probabilities in bootstrap distributions to find confidence intervals and in randomization distributions to test hypotheses. 8. Use the sampling distributions of sample proportions and sample means to answer appropriate questions. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations.	1. Describe how a randomization distribution is generated assuming H_0 is true, and identify its center as the null value. (CO3) 2. Compare sampling, bootstrap, and randomization distributions by their center and purpose. (CO3, CO8) 3. Compute a p-value from a randomization distribution as a proportion of simulated statistics, choosing the correct tail(s) from H_a . (CO3, CO10) 4. Interpret a p-value in context and relate its size to the strength of evidence against H_0 . (CO10) 5. Compare the p-value to α , decide whether to reject H_0 , write a two-step conclusion in context, and avoid saying “accept H_0 ”. (CO10)	e-Text: 4.2 Measuring Evidence with P-Values	MO1, MO2, MO3, MO4	Participation: Randomization Distributions – StatKey: dolphin study (16 flips) and tea vs. coffee re-randomization; count how many simulated statistics are as extreme as observed	MO1, MO2, MO3	Online Homework: 4.2 Measuring Evidence with P-Values (due Sun Oct 25, 11:59 PM)	MO1, MO2, MO3, MO4
		Interactive Lecture: 4.2 Measuring Evidence with P-Values	MO1, MO2, MO3, MO4			Online Homework: 4.3 Determining Statistical Significance (due Sun Oct 25, 11:59 PM)	
		Slides: 4.2 Measuring Evidence with P-Values	MO1, MO2, MO3, MO4	Participation: P-values and Conclusions – resveratrol examples ($p = 0.04, 0.12, 0.0003$): strength of evidence, decision at α , conclusion in context	MO4, MO5	Take-Home Quiz 5 (due Tue Oct 27, 11:59 PM) – covers 3.4, 4.1–4.3	All
		e-Text: 4.3 Determining Statistical Significance	MO4, MO5				
		Interactive Lecture: 4.3 Determining Statistical Significance	MO4, MO5				
		Slides: 4.3 Determining Statistical Significance (includes “What is a p-value?” video)	MO4, MO5				
		StatKey – Randomization Hypothesis Tests	MO1, MO3				